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Batch: C-3 / DV Fri

TITLE: To explore data using filters, joins, and calculations

Objective:

- Apply basic and advanced filters to slice and explore data
- Join multiple tables using relationships in Power BI
- Create calculated columns and DAX measures for custom analysis
- Combine filters, joins, and calculations to answer business questions

Expected outcome of Experiment:

Students will be able to explore and analyze sales data in Power BI by applying filters to isolate specific data subsets, joining multiple tables to enrich the dataset, and creating DAX calculations to derive meaningful business metrics. By the end of the experiment, students will have built an interactive dashboard that demonstrates all three concepts working together

CO3

Apply data visualization best practices

Books/ Journals/ Websites referred:

Background Theory:

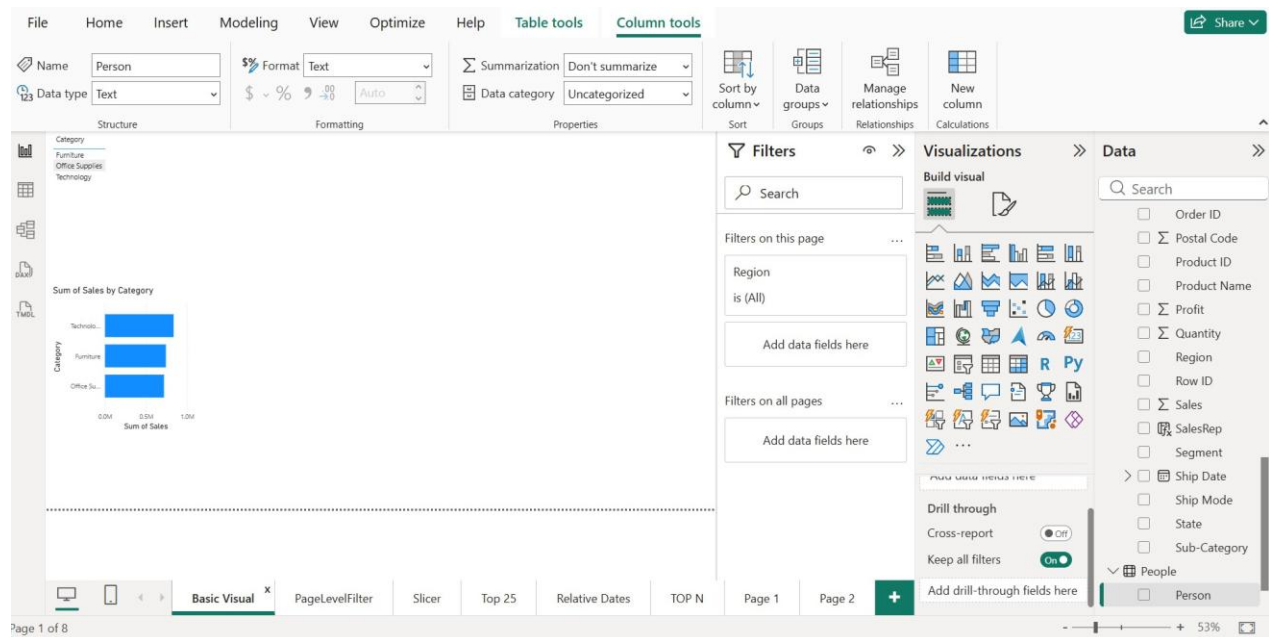
A filter is a condition applied to data that restricts what is displayed in a visual or report. Filtering does not delete data — it simply hides rows that do not match the specified criteria. When you remove the filter, all data becomes visible again. Filters are used to focus analysis on a specific subset of data, such as a particular region, time period, or product category. In everyday business scenarios, filters answer questions like: 'Show me only sales from the West region' or 'Display only orders placed in 2021.' Filters are the most basic but most frequently used tool in any BI platform.

A join is an operation that combines rows from two or more tables based on a related column. In relational databases, tables are deliberately kept separate and lean — each table stores one type of information. For example, an Orders table stores transaction data, while a Customers table stores customer details. A join links these two tables through a shared key (such as Customer ID) so that combined analysis is possible. Without joins, analysts would be forced to work with a single massive flat table containing all columns, which leads to data redundancy, inconsistency, and poor performance. Joins are the backbone of relational data modeling.

DAX stands for Data Analysis Expressions. It is the formula language used in Power BI (as well as Excel Power Pivot and Azure Analysis Services) to define custom calculations. DAX is designed specifically for working with relational data models and is highly optimized for aggregation, filtering, and time intelligence operations. DAX syntax is similar to Excel formulas in style but operates at the table and column level rather than the cell level. A single DAX measure can aggregate millions of rows in milliseconds and dynamically recalculate based on the current filter context.

Note: Perform below task in power Bi on sample supermarket dataset.

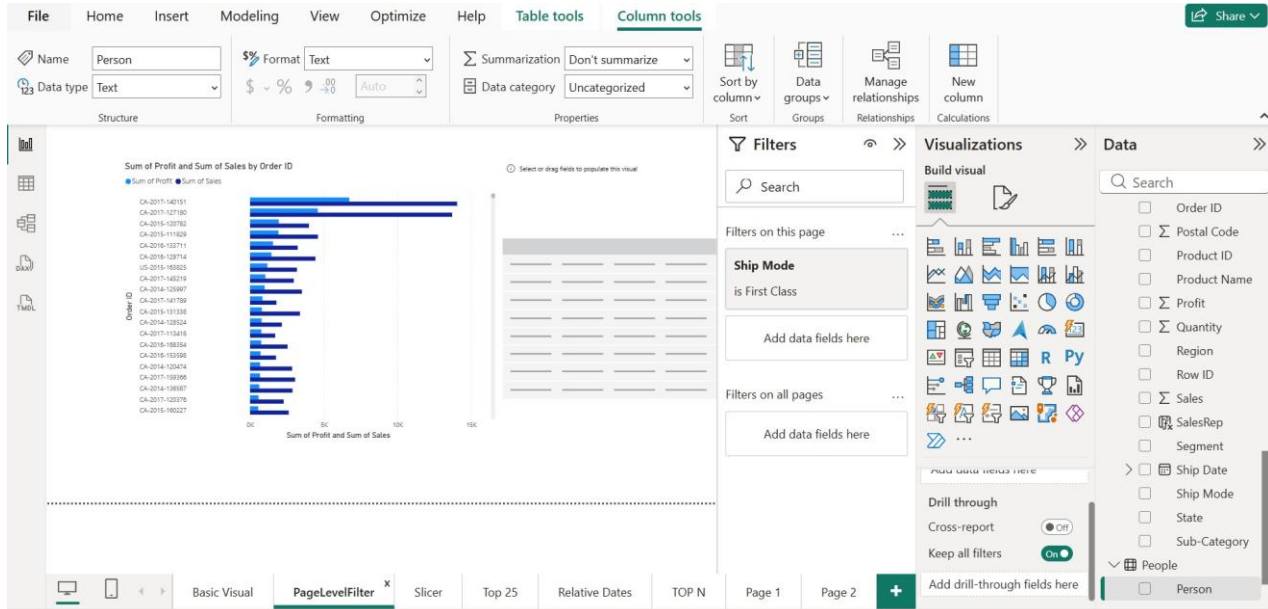
Task 1: Apply a Basic Visual-Level Filter



Steps:

1. Create a Bar Chart showing Sum of Sales by Category.
2. In the Filters pane, expand 'Filters on this visual'.
3. Drag the Region field into the filter area.
4. Select only 'West' from the filter values.
5. Observe how only West region data now appears in the chart.

Task 2: Apply a Page-Level Filter

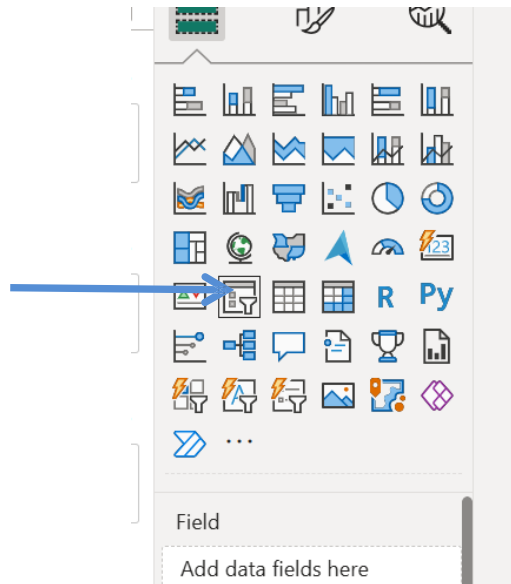


Steps:

6. Add a second visual to the same page — a table showing Order ID, Sales, and Profit.
7. In the Filters pane, go to 'Filters on this page'.
8. Drag Ship Mode into this section.
9. Select 'First Class' only.
10. Observe that BOTH visuals on the page now show First Class data only.

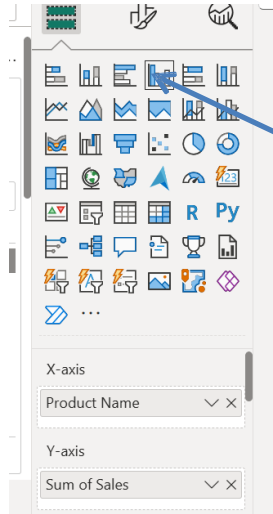
Task 3: Create an Interactive Slicer

Steps:



11. Click on an empty area of the canvas.
12. In the Visualizations pane, select the Slicer visual.
13. Drag the Category field into the slicer.
14. Create a second slicer using the Year from Order Date (use the date hierarchy).
15. Test the slicers by selecting different categories and years.

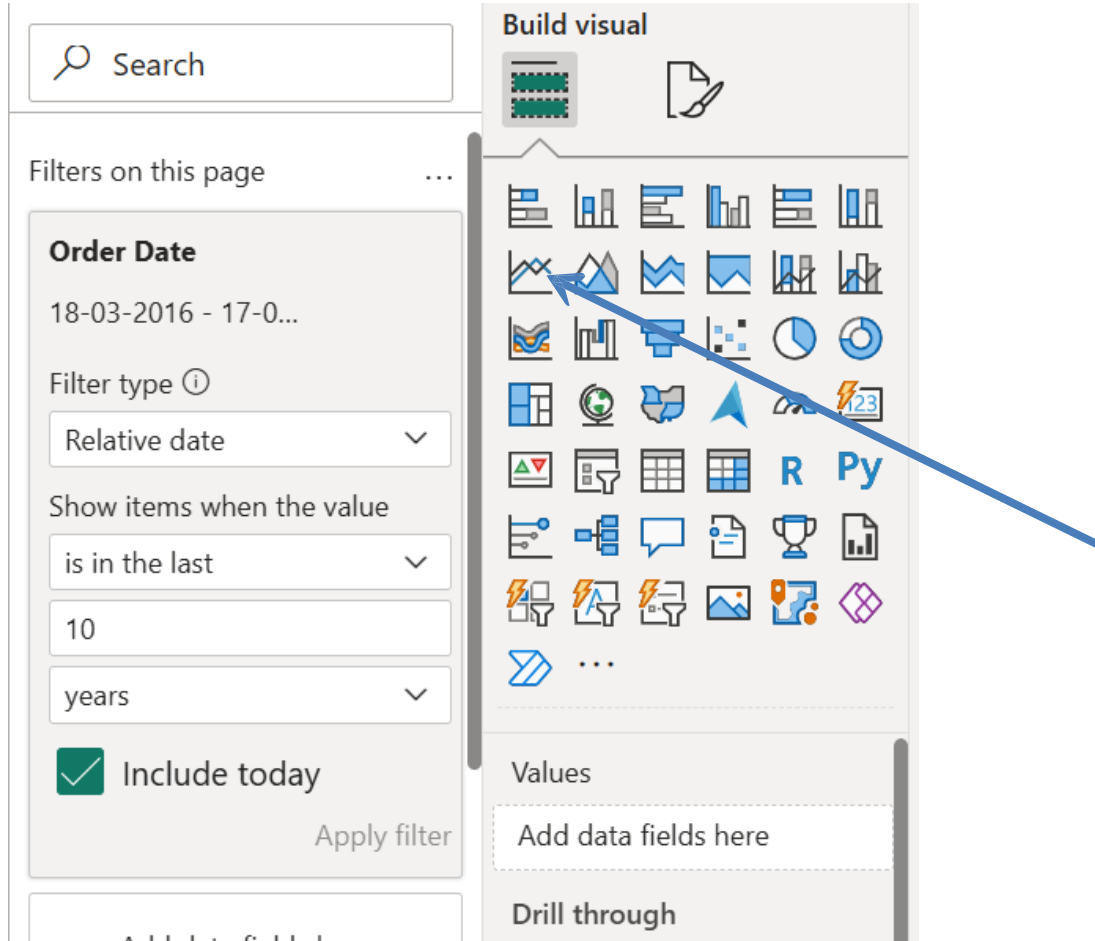
Task 4: Use Top N Filter



Steps:

16. Create a Bar Chart showing Sum of Sales by Product Name.
X axis: product name and Y axis : Sum Sales
17. In Filters on this visual, drag Product Name into the filter area.
In filter add product name
18. Change filter type from 'Basic filtering' to 'Top N'.
19. Set Show items = Top 10, By value = Sum of Sales.
20. Click Apply filter.

Task 5: Apply a Date Range Filter



The screenshot shows the Power BI interface. On the left, the 'Filters on this page' pane is open, showing a filter for 'Order Date' with a date range of '18-03-2016 - 17-0...'. The filter type is set to 'Relative date', and the value is '10 years'. The 'Include today' checkbox is checked. On the right, the 'Build visual' pane is open, showing a grid of visualization icons. A blue arrow points from the 'Line chart' icon in the 'Build visual' pane to the 'Order Date' filter in the 'Filters on this page' pane.

Steps:

21. Create a Line Chart showing Sum of Sales over Order Date.
22. In Filters on this visual, drag Order Date into the filter area.
23. Change filter type to 'Relative date'.
24. Set it to show 'Last 2 Years'.
25. Alternatively, use 'Between' to set a custom date range (e.g., Jan 2020 to Dec 2021).

Task 6: View and Understand Existing Relationships

Steps:

26. Click the Model view icon on the left sidebar (looks like three connected shapes).
27. Observe the relationship lines between Orders, Returns, and People tables.
28. Double-click on the relationship line between Orders and Returns.
29. Note the joining key (Order ID), cardinality, and cross-filter direction.
30. Close the dialog. Sketch or note down what you observe.

Task 7: Create a New Relationship Manually

Steps:

31. In Model view, check if Orders and People tables are connected.
32. If not, drag the Region field from the People table to the Region field in the Orders table.
33. In the relationship dialog, confirm Cardinality = Many to One, Cross-filter = Single.
34. Click OK to create the relationship.
35. Go back to Report view and create a table showing Region, Sales Rep (from People), and Sum of Sales.

Task 8: Identify Returned Orders Using a Join

Steps:

36. Go to Home > Transform Data to open Power Query Editor.
37. Select the Orders query in the left panel.
38. Go to Home > Merge Queries.
39. Select Returns as the second table, match on Order ID, and choose Left Outer join.
40. Expand the merged column and bring in the 'Returned' column.
41. Close and Apply. Back in Power BI, create a table showing Order ID, Sales, and Returned status.

Use maximize view of power bi query editor

Added new columns in the new table which gets automatically added in to power bi

Task 9: Calculate Total Sales for Returned vs Non-Returned Orders

Steps:

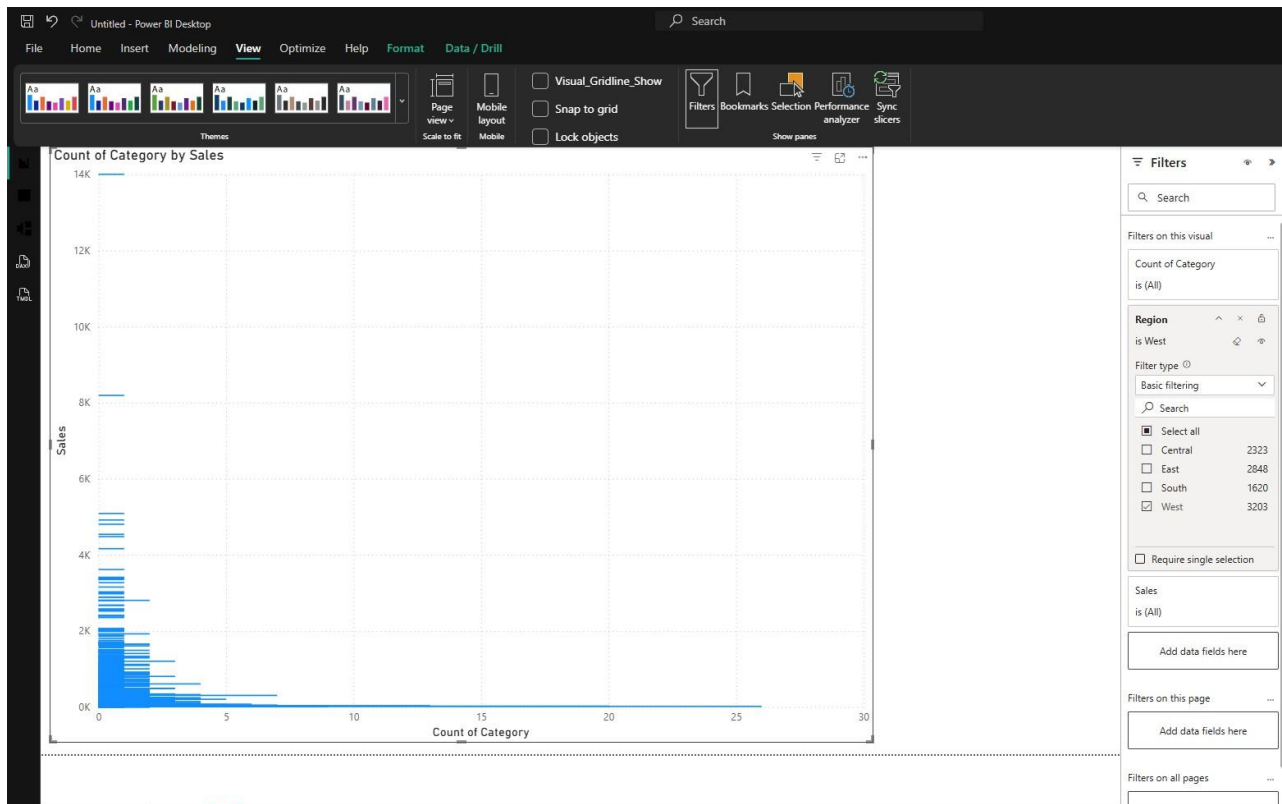
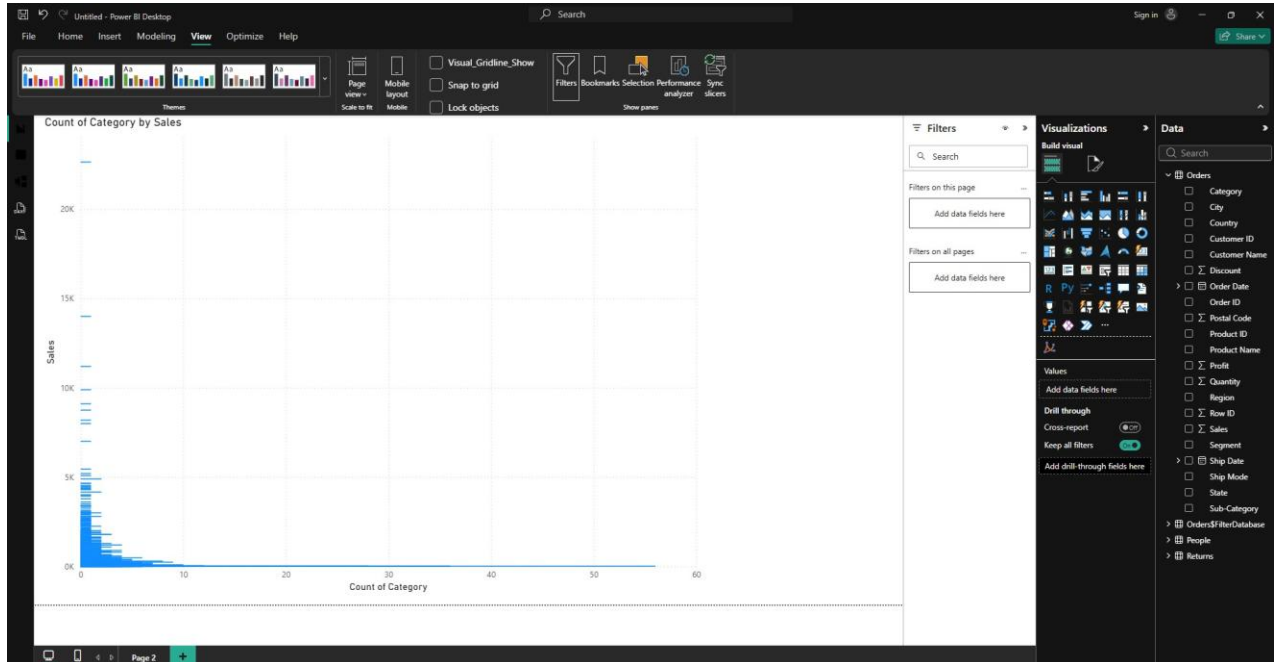
42. Using the Returned column you added in Task 8, create a Bar Chart.
43. Set the Y axis to Returned (Yes / blank) and X axis to Sum of Sales.
44. Add a data label to show the actual sales values.
45. Format the chart with appropriate title: 'Sales: Returned vs Non-Returned'.

Task 10: Use RELATED Function Across Tables

Steps:

46. In the Report view, go to the Orders table in the Data pane.
47. Right-click > New Column.
48. Enter the formula: Sales Rep = RELATED(People[Person])
49. This pulls the Sales Rep name from the People table into each Order row.
50. Create a table visual showing Order ID, Customer Name, Sales, and Sales Rep.

Task 1: Apply a Basic Visual-Level Filter

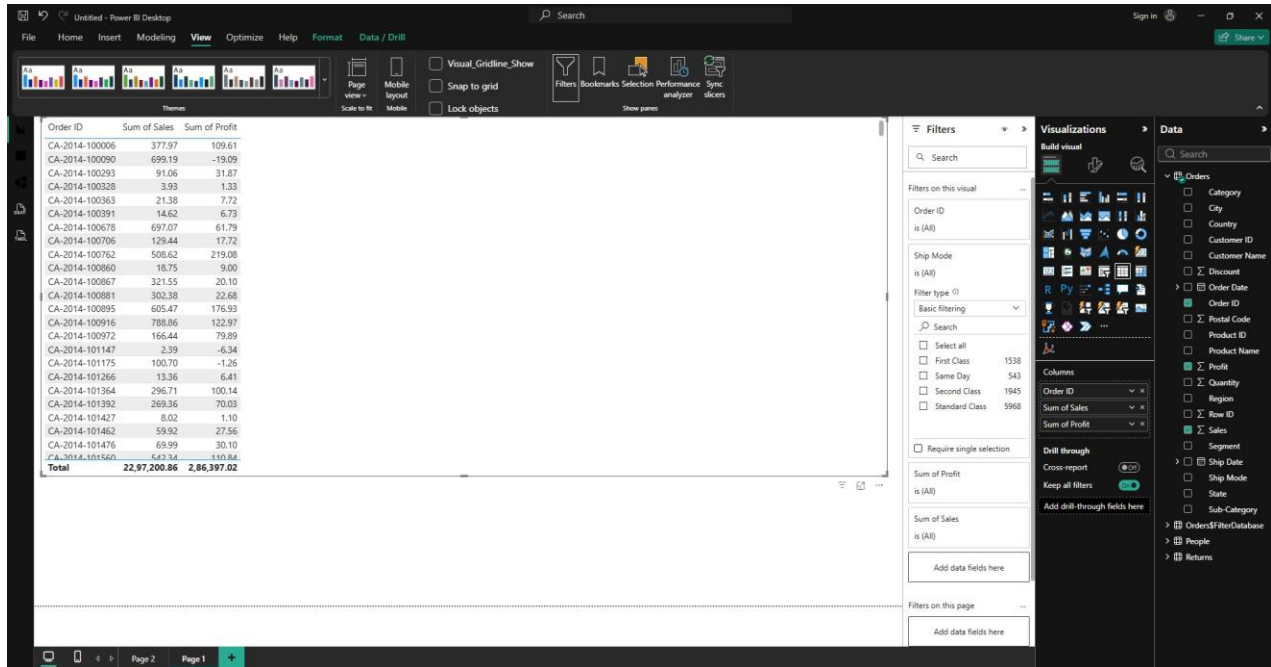


Result:

The bar chart initially displayed sales for all regions. After applying the visual-level filter for the *West*

region, the chart updated to show only the sales data corresponding to the West region. This demonstrates how visual-level filters restrict data for a specific visual without affecting others.

Task 2: Apply a Page-Level Filter

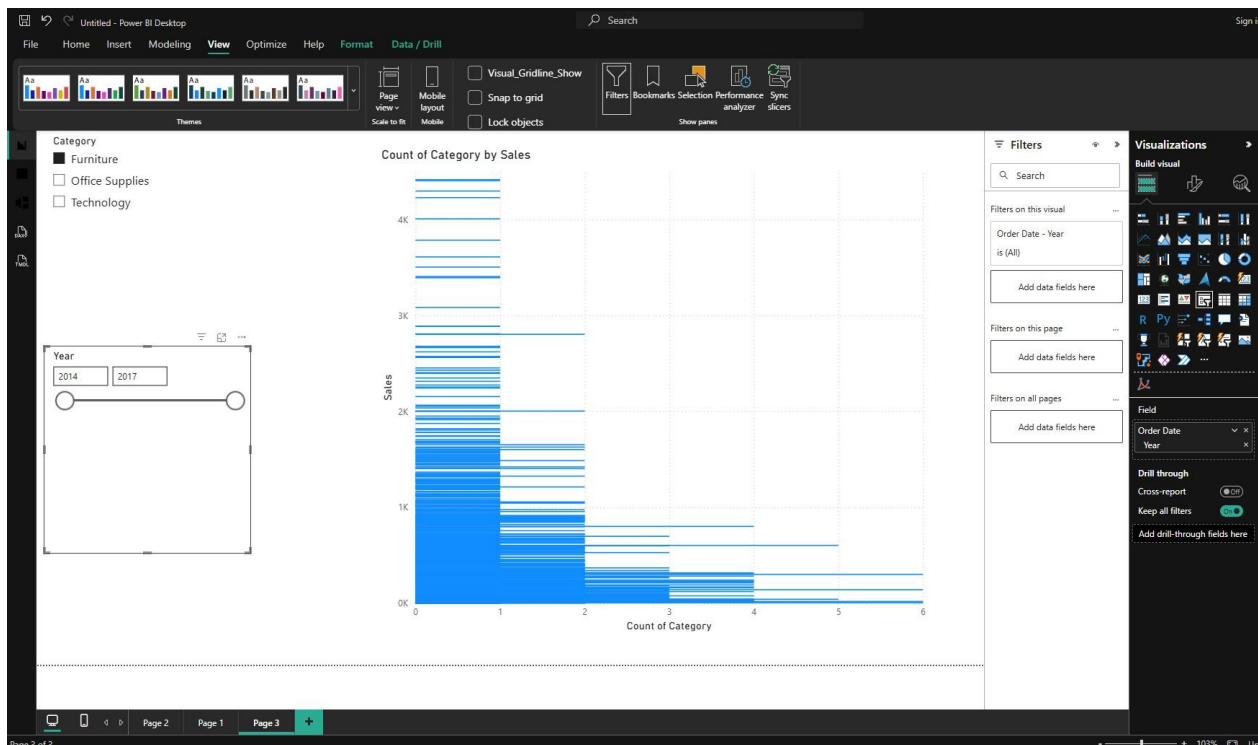
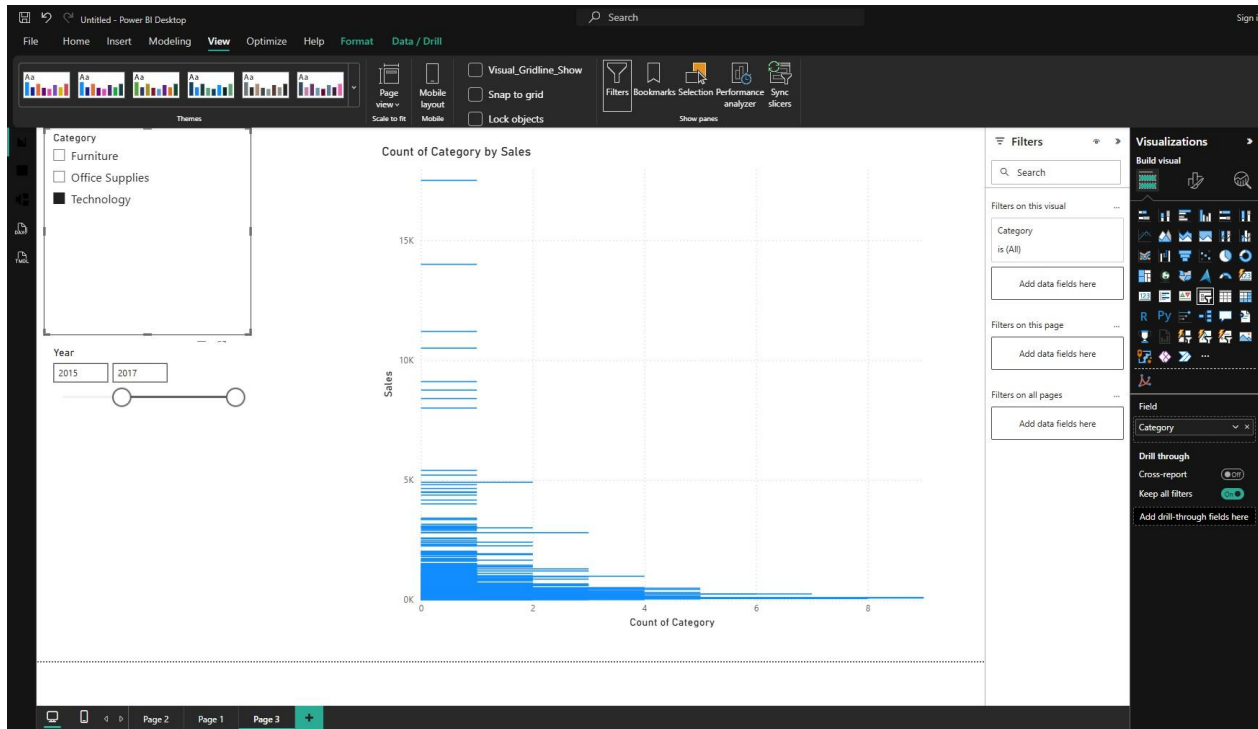


Result:

After applying the page-level filter for *First Class* ship mode, both the bar chart and the table visual were

updated simultaneously. Only orders with First Class shipping were displayed, showing how page-level filters affect all visuals on the page.

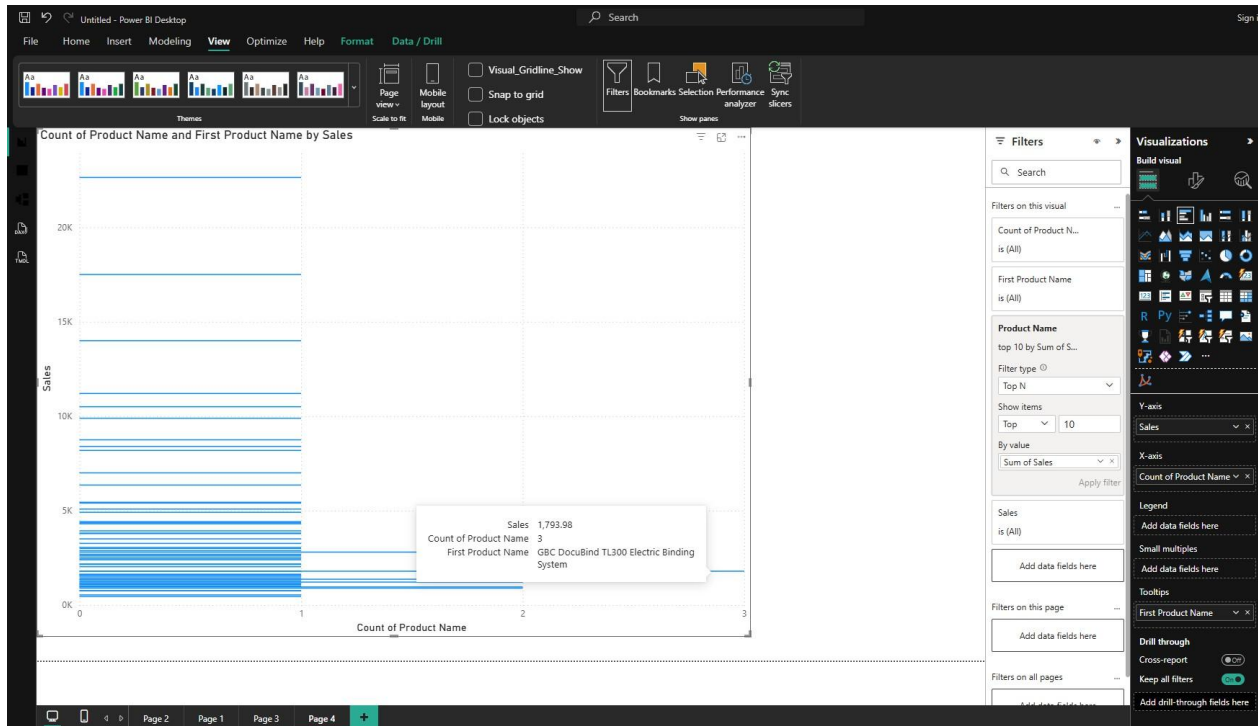
Task 3: Create an Interactive Slicer



Result:

Slicers for *Category* and *Year* were created successfully. Selecting different values dynamically updated all visuals on the report. This provided an interactive way to explore data across different categories and time periods.

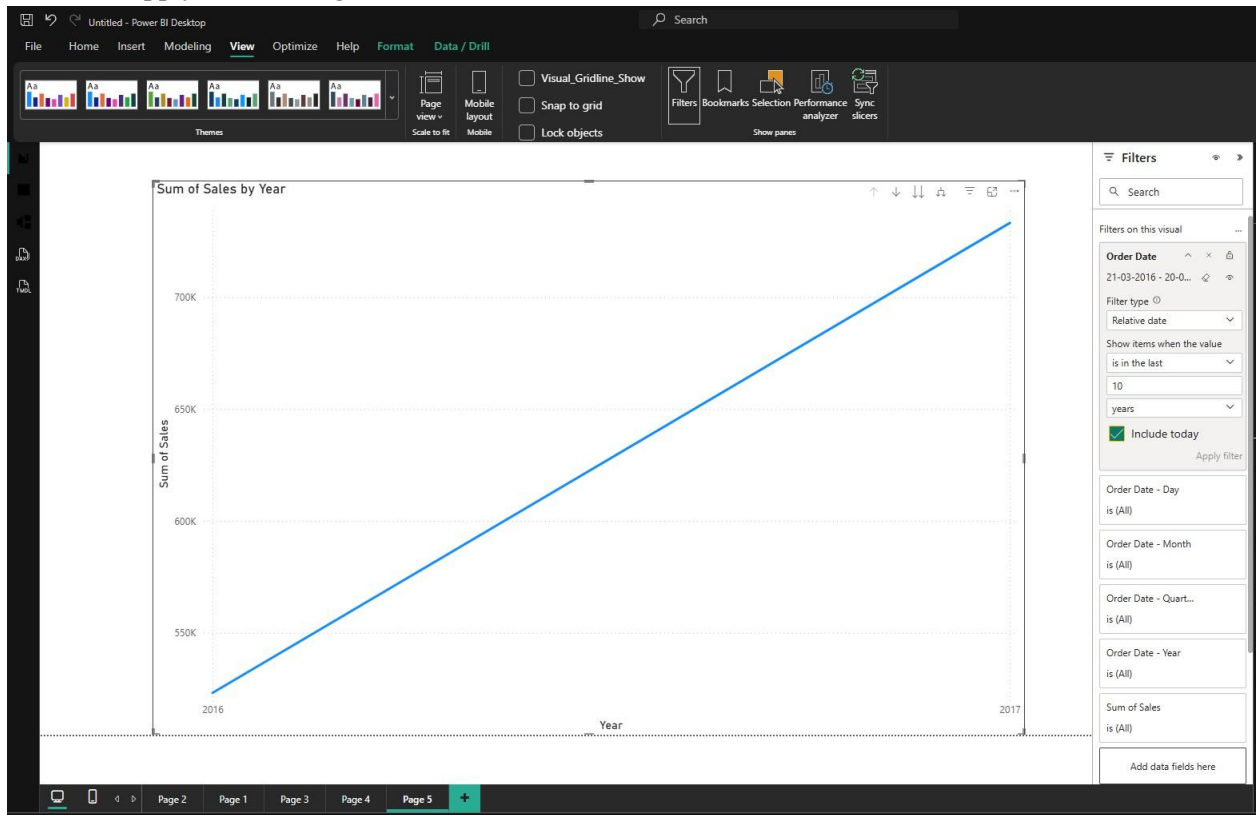
Task 4: Use Top N Filter



Result:

The bar chart displayed only the *Top 10 products* based on total sales after applying the Top N filter. This helped in identifying the highest-performing products quickly without clutter from less significant data.

Task 5: Apply a Date Range Filter

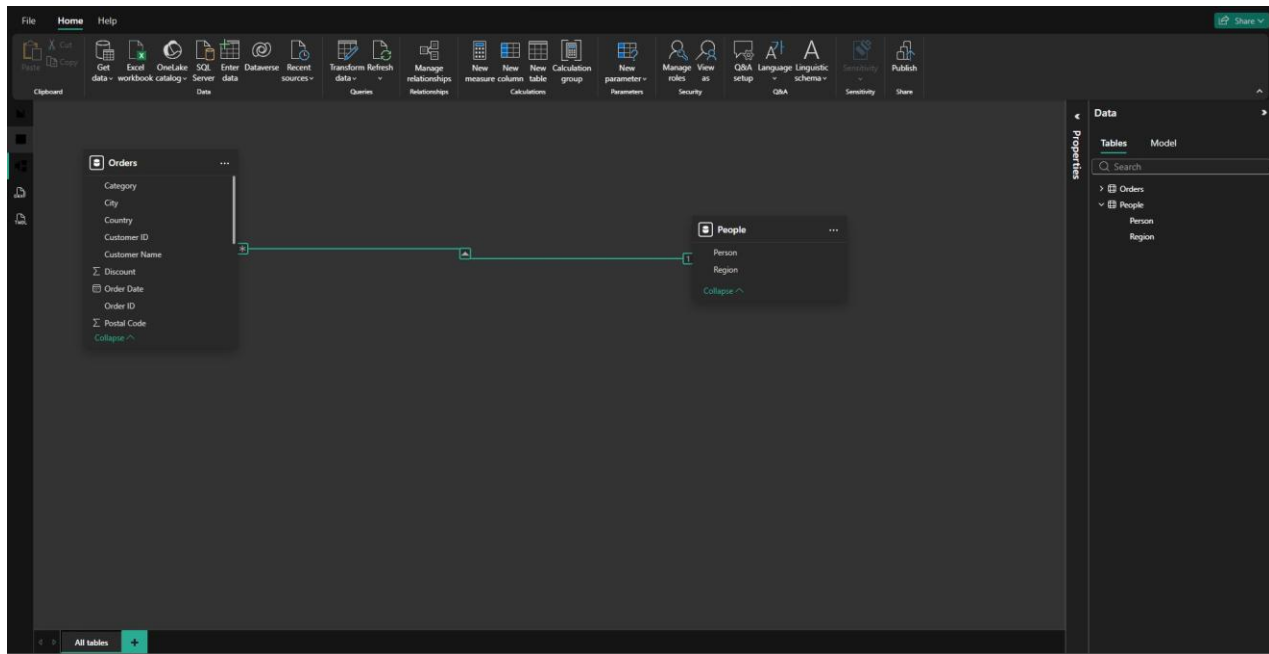


Result:

The line chart showed sales trends over time. After applying the relative date filter (Last 2 Years), the chart updated to display only recent data. This allowed focused analysis on recent performance trends.

Task 6: View and Understand Existing Relationships

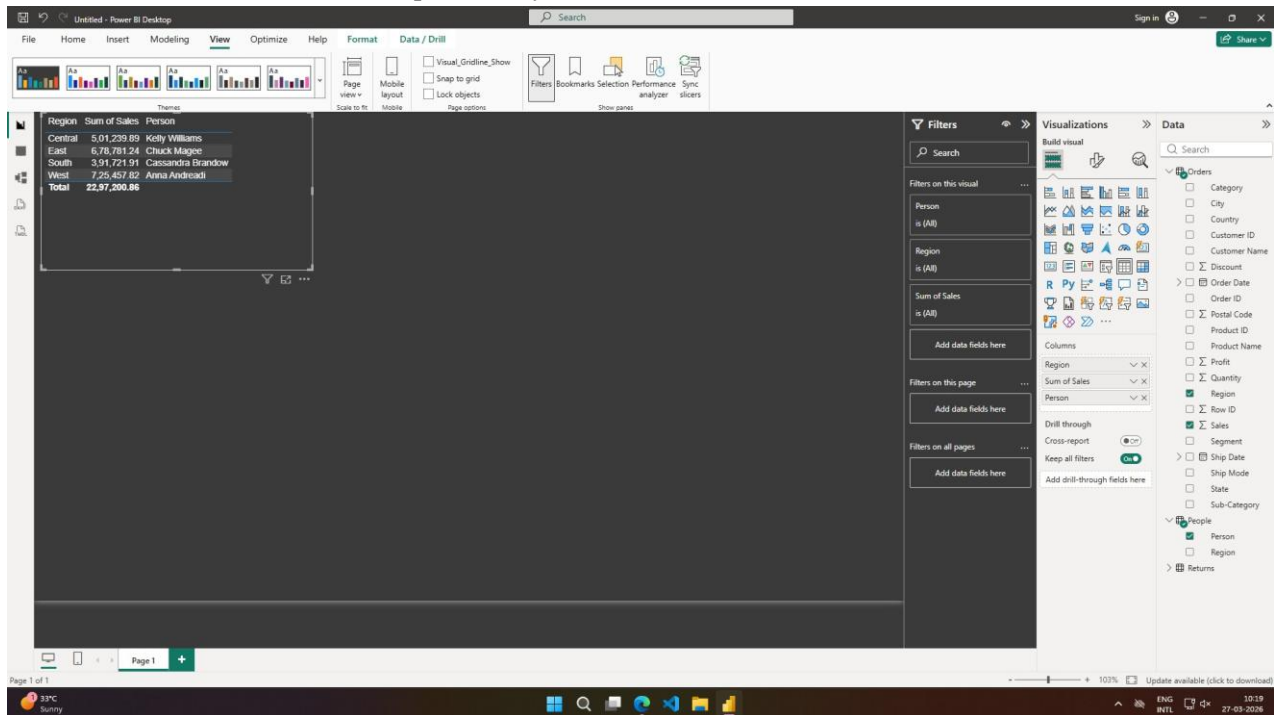
Sub-Category	Product Name	Sales	Quantity	Discount	Profits	Returns
Bookcases	Bush Somerset Collection Bookcase	261.96	2	0	41.9336	Table
Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	3	0	219.582	Table
Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2	0	6.8724	Table
Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.031	Table
Storage	Eldon Fold 'N Roll Cart System	22.368	2	0.2	2.5354	Table
Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	7	0	14.1694	Table
Art	Newell 322	7.28	4	0	1.9656	Table
Phones	Milco 5320 IP Phone VoIP phone	907.152	6	0.2	90.7252	Table
Binders	DAL Angle-View Binders with Locking Rings by Small	18.504	3	0.2	5.7825	Table
Appliances	Belkin PSC200VTEL 6 Outlet Surge	124.9	5	0	34.47	Table
Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2	85.3092	Table
Phones	Konftel 250 Conference phone - Charcoal black	912.424	4	0.2	68.3568	Table
Paper	Xerox 1967	15.552	3	0.2	5.4452	Table
Binders	Fellowes P5200 Plastic Comb Binding Machine	407.976	3	0.2	132.2352	Table
Appliances	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HL...	68.81	5	0.8	-123.854	Table
Binders	Stamps DuraTech Recycled Plastic Frosted Binders	2.544	3	0.8	-3.436	Table
Storage	Stor-O-Stor Shelving, Vertical 5-Shelf 72"W x 36"W x 18 1/2"D	663.48	6	0	13.3276	Table
Storage	Fellowes Super Stor/Drawer	55.5	2	0	9.99	Table
Art	Newell 341	8.56	2	0	2.4824	Table
Phones	Cisco SPA 501G IP Phone	213.48	3	0.2	16.011	Table
Binders	Wilson Jones Hanging View Binder, White, 1"	22.72	4	0.2	7.384	Table
Art	Newell 318	19.46	7	0	5.0896	Table
Appliances	Acco-Six Outlet Power Strip, 4' Cord Length	69.34	7	0	15.6884	Table
Chairs	Global Deluxe Stacking Chair, Gray	71.372	2	0.3	-1.0396	Table
Tables	Bretford CR4500 Series Slim Rectangular Table	1044.63	3	0	240.2649	Table
Binders	Wilson Jones Active Use Binders	11.648	2	0.2	4.2224	Table
Accessories	Ination RGB Mini TravelDrive USB 2.0 Flash Drive	90.57	3	0	11.7741	Table
Bookcases	Riverside Palati Royal Lawyers Bookcase, Royale Cherry Finish	3083.43	7	0.5	-1865.0522	Table
Binders	Auray Recycled Fleece View Covers for Binding Systems	9.034	2	0.7	-7.0512	Table
Furnishings	Haworth Miller 13 1/4" Diameter Brushed Chrome Round Wall Clock	124.2	3	0.2	13.525	Table
Envelopes	Poly String Tie Envelopes	3.264	2	0.2	1.3036	Table
Art	BOSTON Model 1800 Electric Pencil Sharpener, Putty/Woodgrain	86.304	6	0.2	9.7892	Table
Binders	Acco Pressboard Covers with Storage Hooks, 14 7/8" x 11", Executive ...	6.858	6	0.7	-5.715	Table
Art	Lumber Crayons	15.76	2	0.2	3.546	Table
Paper	Easy-staple paper	29.472	3	0.2	9.9468	Table
Phones	GE 39524614	1097.544	7	0.2	123.4737	Table



Result:

In the Model view, relationships between Orders, Returns, and People tables were observed. The Orders and Returns tables were connected using *Order ID*. The relationship showed proper cardinality and cross-filter direction, helping understand how data flows between tables.

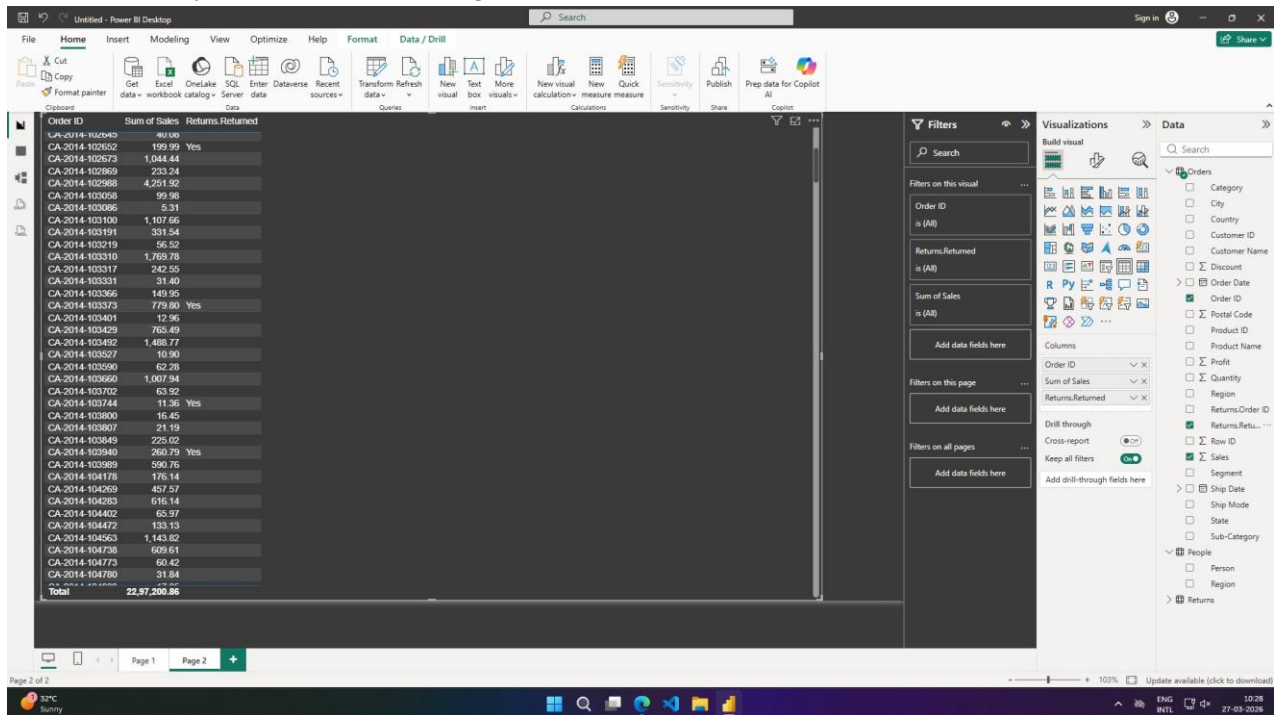
Task 7: Create a New Relationship Manually



Result:

A new relationship between Orders and People tables was created using the *Region* field. After establishing this relationship, a table visual successfully displayed Region, Sales Rep, and Sales, confirming that data from both tables was correctly linked.

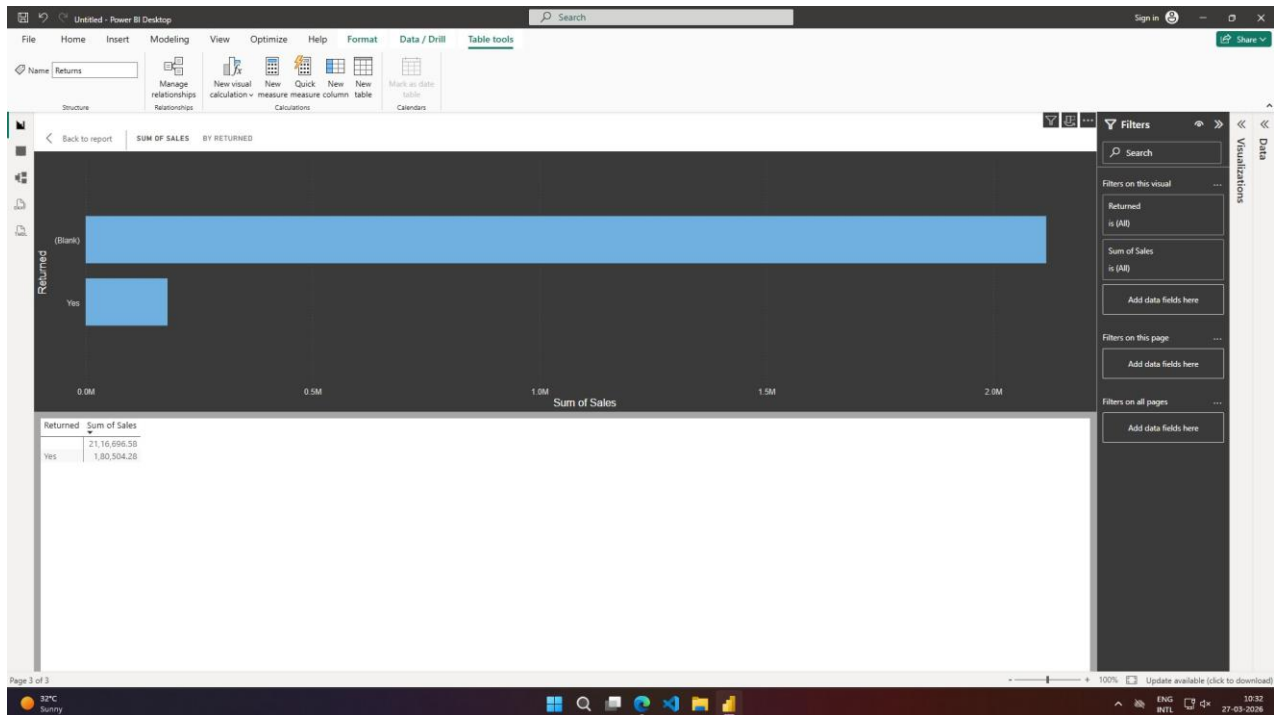
Task 8: Identify Returned Orders Using a Join



Result:

Using Power Query, a Left Outer Join was performed between Orders and Returns tables based on Order ID. A new column *Returned* was added, indicating whether an order was returned or not. This enriched the dataset with return status information.

Task 9: Calculate Total Sales for Returned vs Non-Returned Orders



Result:

A bar chart was created to compare sales for returned and non-returned orders. It clearly showed the difference in sales values between the two categories, helping analyze the impact of returns on overall sales.

Conclusion:

Through this experiment, we learned how to effectively explore and analyze data in Power BI using filters, joins, and DAX calculations. Filters helped in focusing on specific subsets of data, while joins enabled combining multiple tables for richer insights. The use of DAX allowed us to create custom calculations and derive meaningful business metrics.

Post Laboratory Questions

- What is the difference between a visual-level filter and a slicer in Power BI?
 - A **Visual-Level Filter** is applied to a single visual and affects only that chart or table.
 - A **Slicer** is an interactive filter placed on the report canvas that can control multiple visuals at once.
 - Visual filters are static (set in filter pane), while slicers are dynamic and user-controlled.

- What is a relationship in Power BI and why is it needed?

A relationship in Power BI is a connection between two tables using a common column (like Order ID or Region).

It is needed because:

- It allows data from multiple tables to be combined
- Enables accurate analysis across tables
- Avoids data redundancy
- Supports functions like `RELATED()` and filtering across tables

- What is the difference between a Calculated Column and a Measure in DAX?

Calculated Column	Measure
Computed row by row	Computed on aggregation
Stored in table	Not stored, calculated on the fly
Used for slicing/filtering	Used in visuals for calculations
Example: adding Sales Rep column	Example: SUM of Sales